Session 14: Sorting, Tables & Referencing in Excel

Presented by

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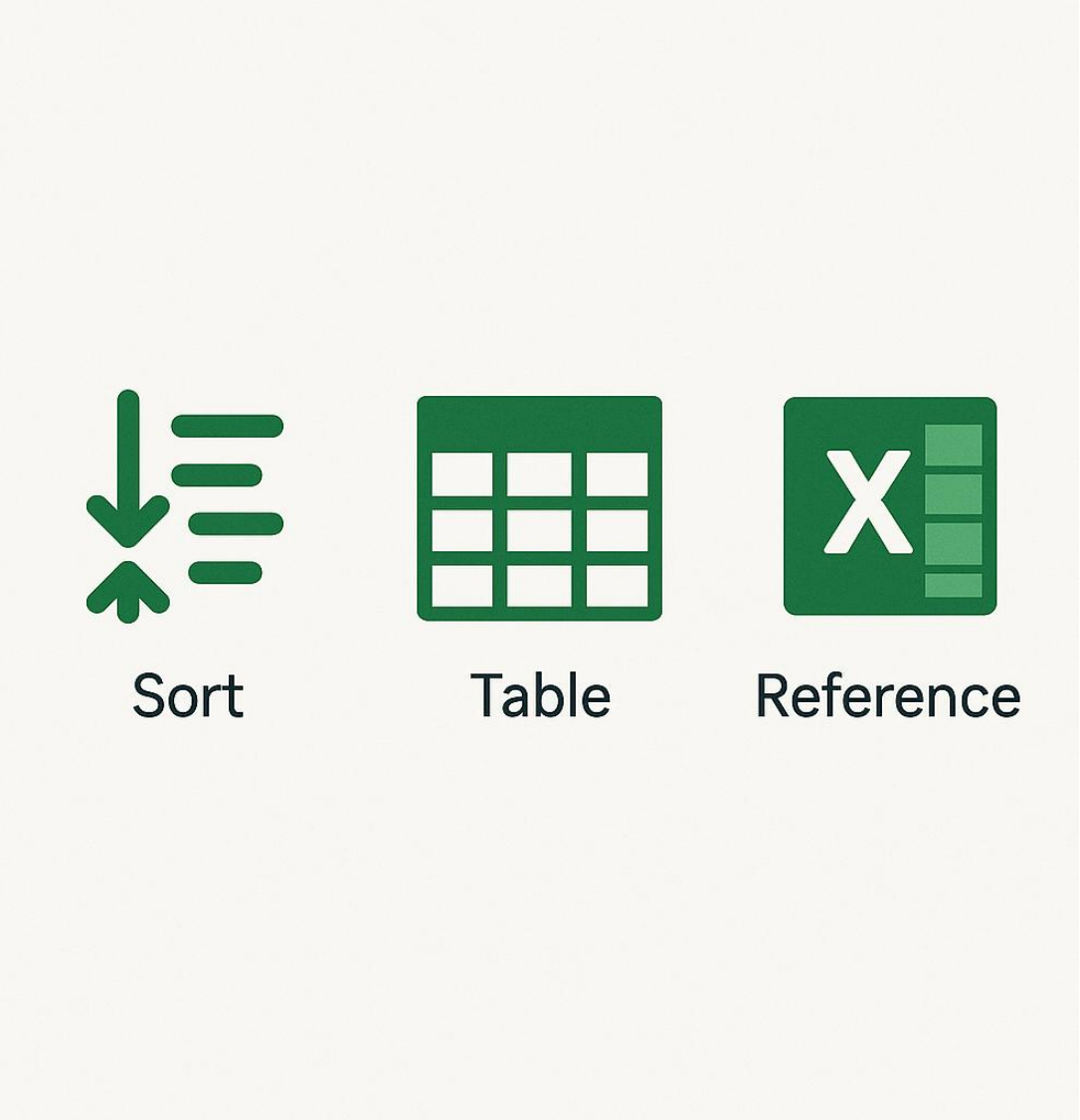
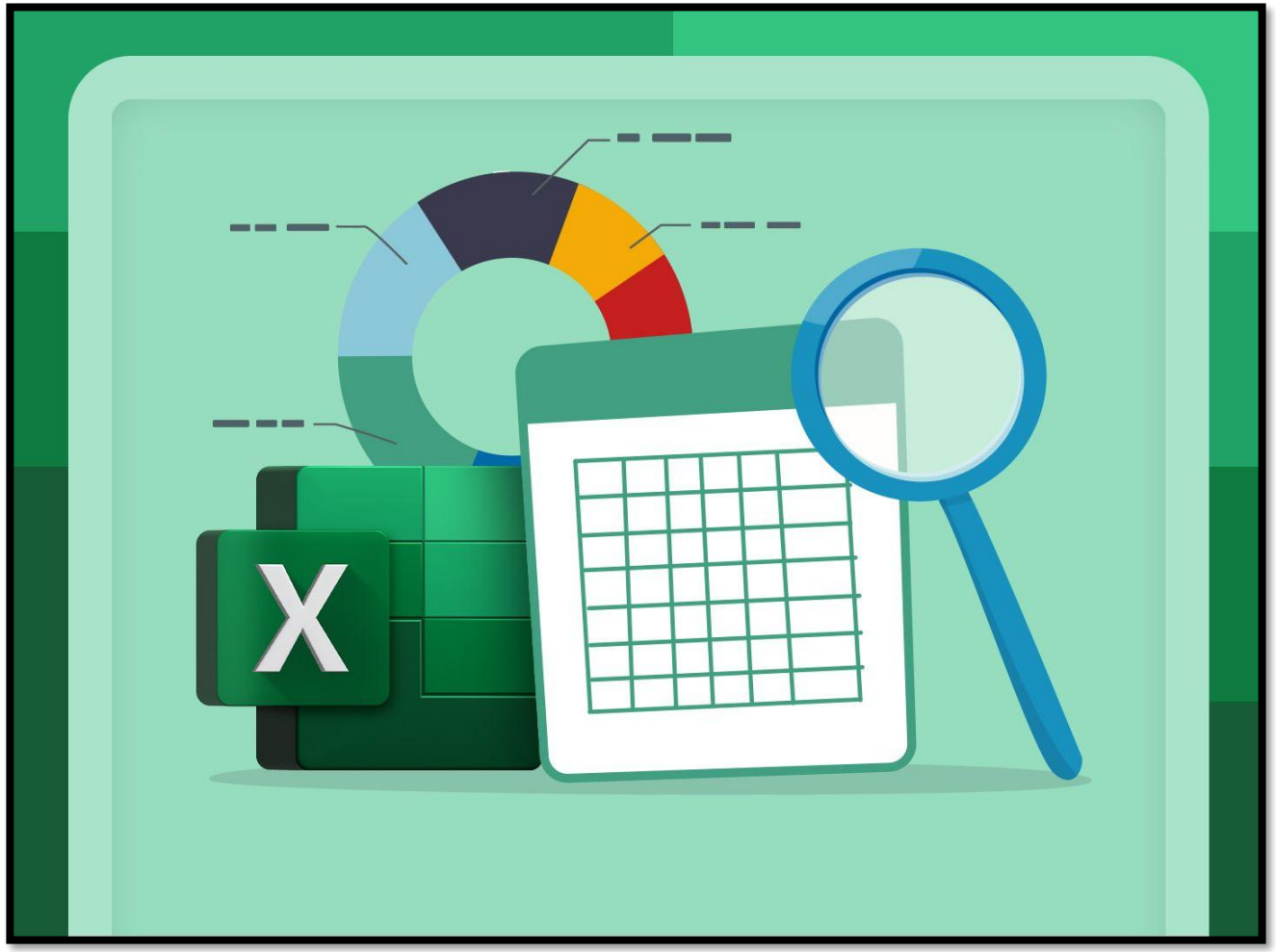
Learning Objectives

- *By the end of this session, you will be able to:*
- *Sort Excel data by value or category*
- *Create and format tables*
- *Apply relative and absolute references in formulas*



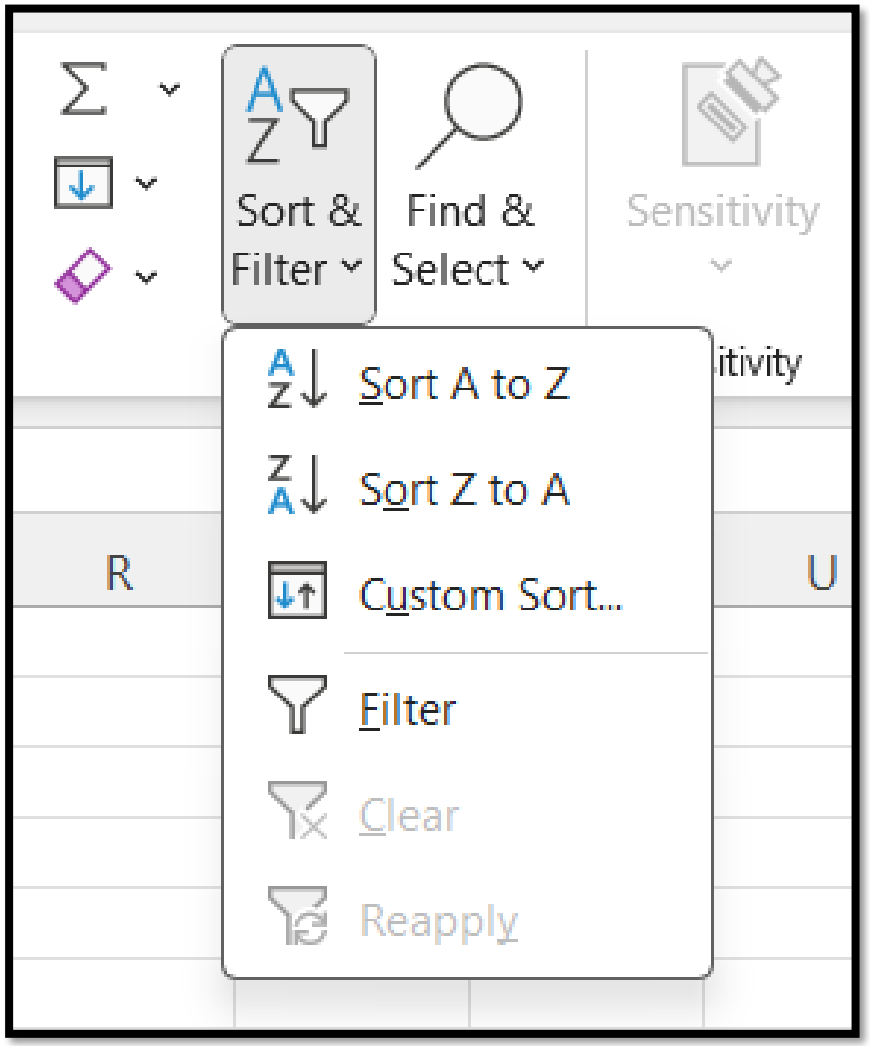
Why Organize Data?

- *Sorting & tables help you find and analyze data easily.*
- *Referencing makes your formulas reusable and efficient.*



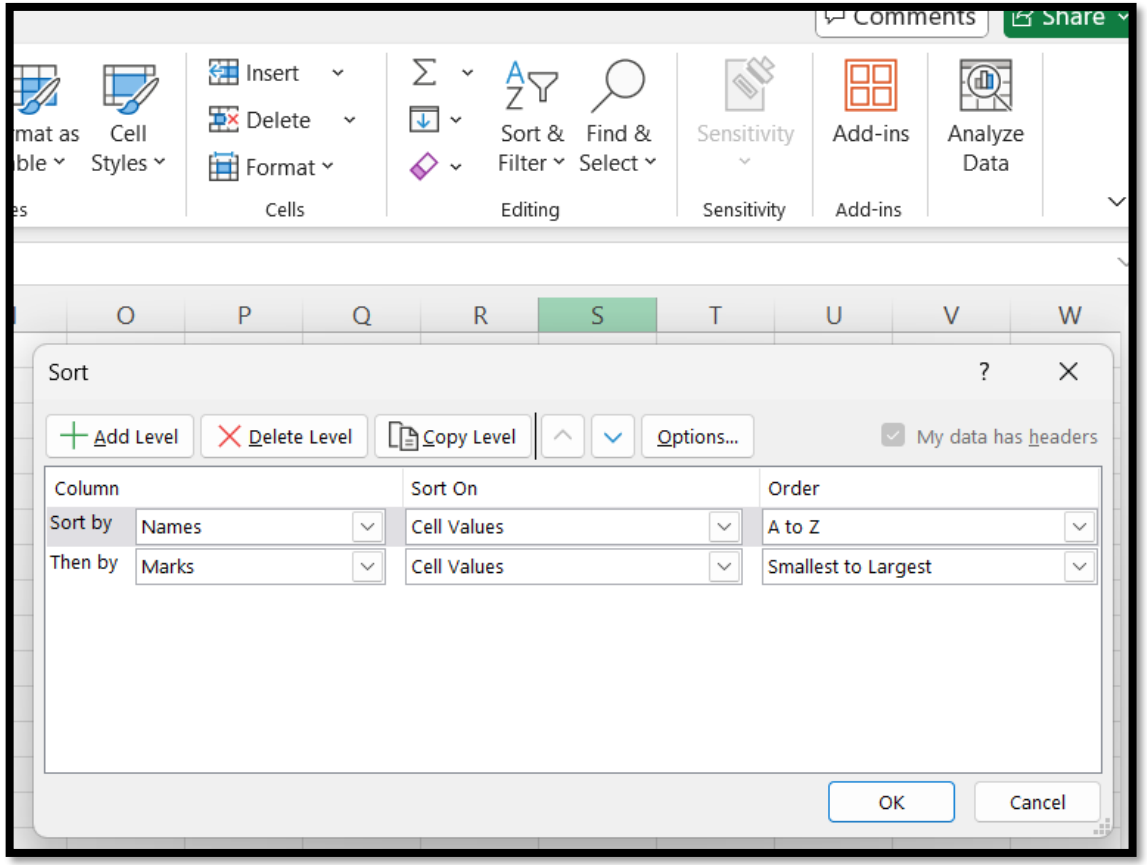
Sorting Data in Excel

- *Sort A-Z / Z-A by text, numbers, dates.*
- *Custom sort with multiple columns.*
- *Use "Sort by Color"*
- *Practical uses: Sorting marks, names, dates.*



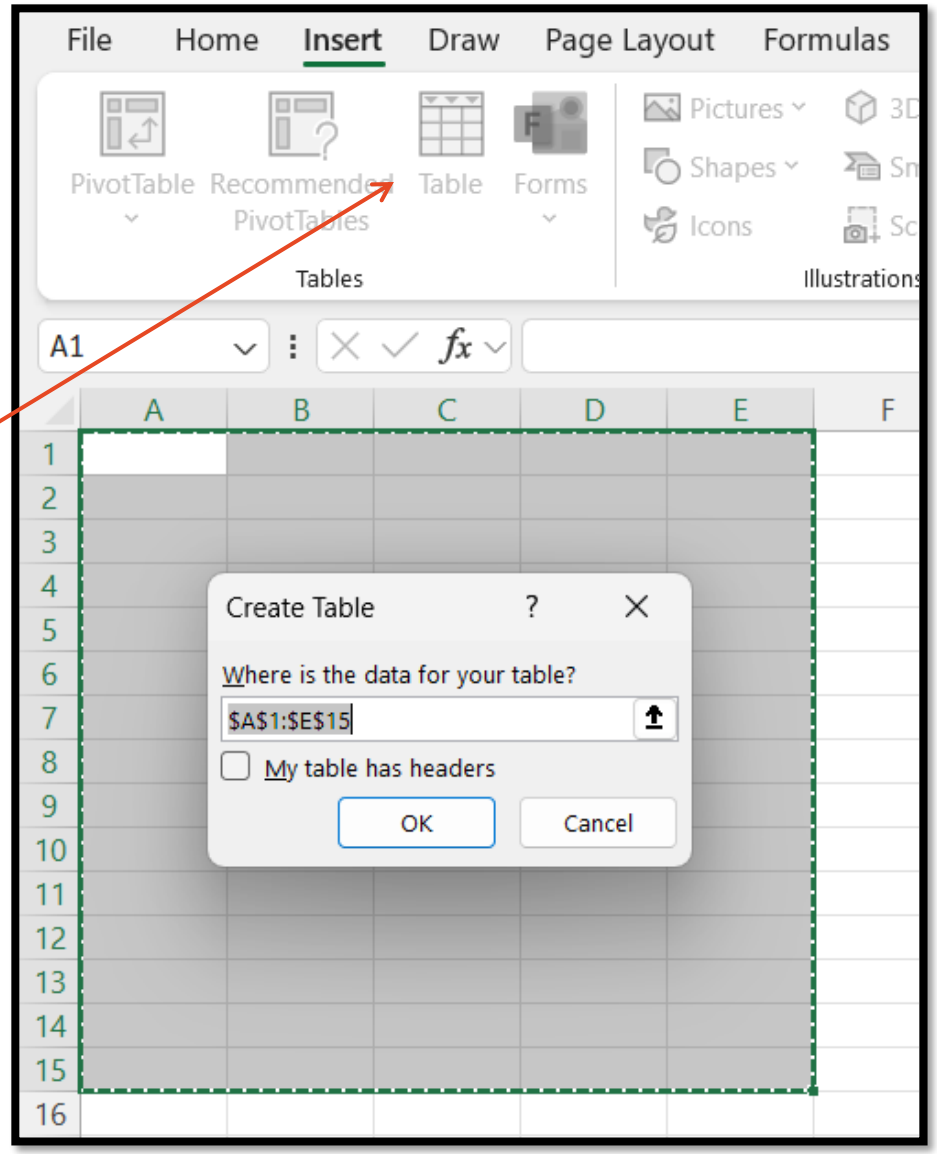
Multi-Level Sorting

- *Sort by multiple columns (e.g., by Subject, then by Marks)*
- *Use “Add Level” in Sort dialog*



Using Tables in Excel

- *Insert → Table to convert data into a structured table.*
- *Tables allow easy filtering, clear formatting, and structured referencing.*
- *Helps in large data management.*



What is Cell Referencing?

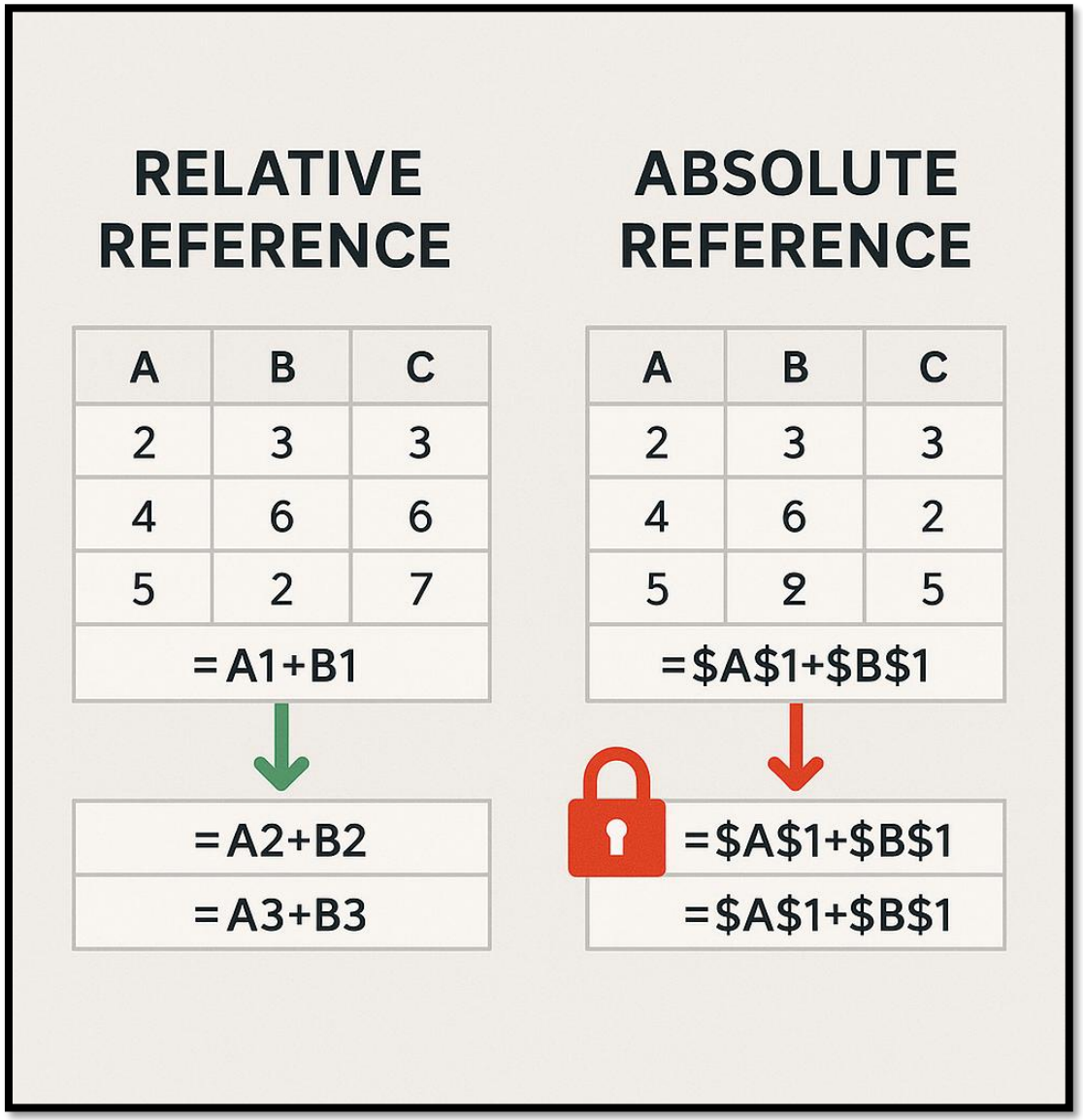
- ✓ *Allows you to use **cell addresses in formulas**.*

Types:

Relative Reference:

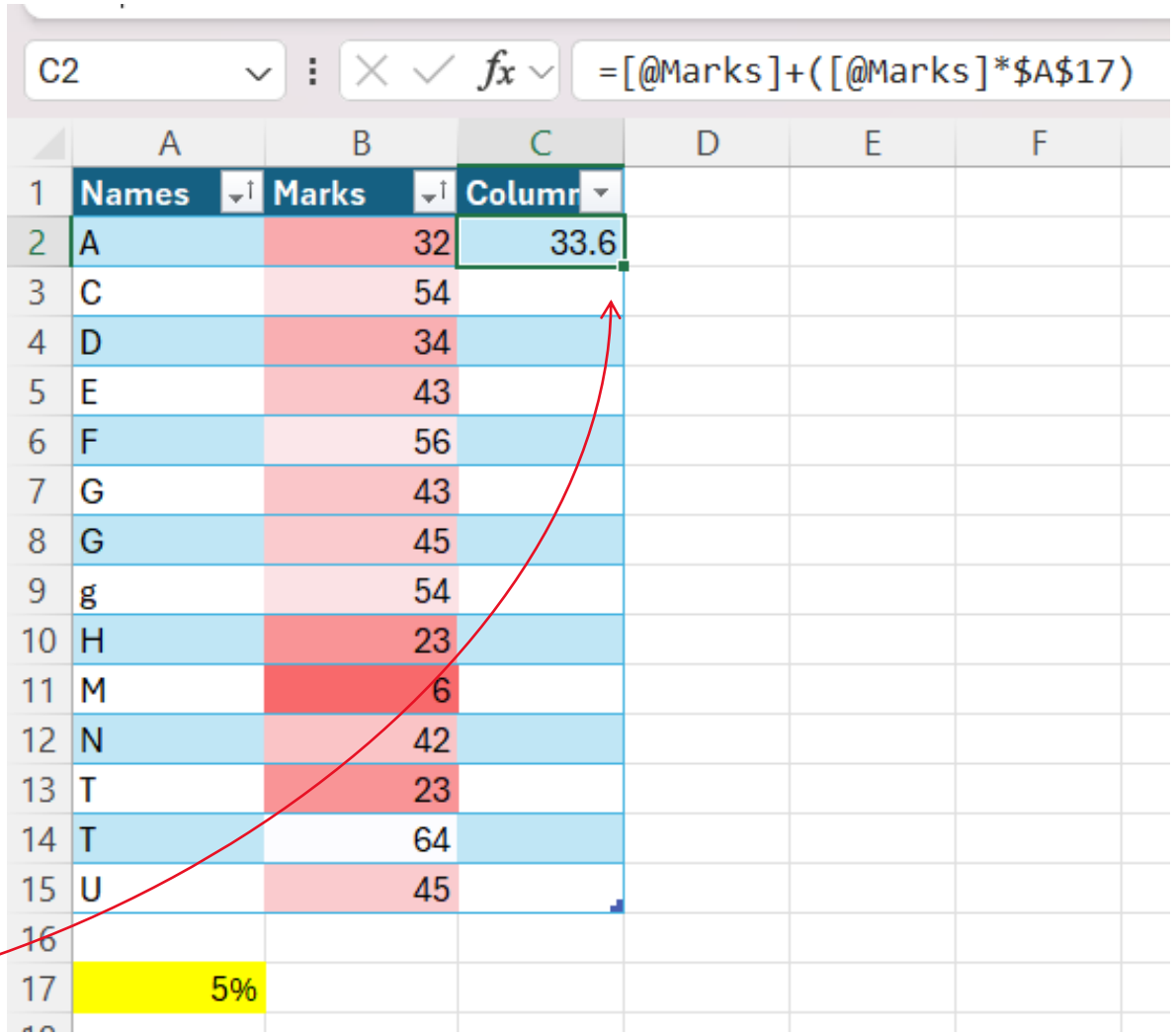
- *Adjusts when copied (A1).*
- *Absolute Reference: Fixed with $ (A1).*
- *Mixed Reference: Partially fixed (A$1, $A1).*

✓ *Makes formulas scalable when copying.*



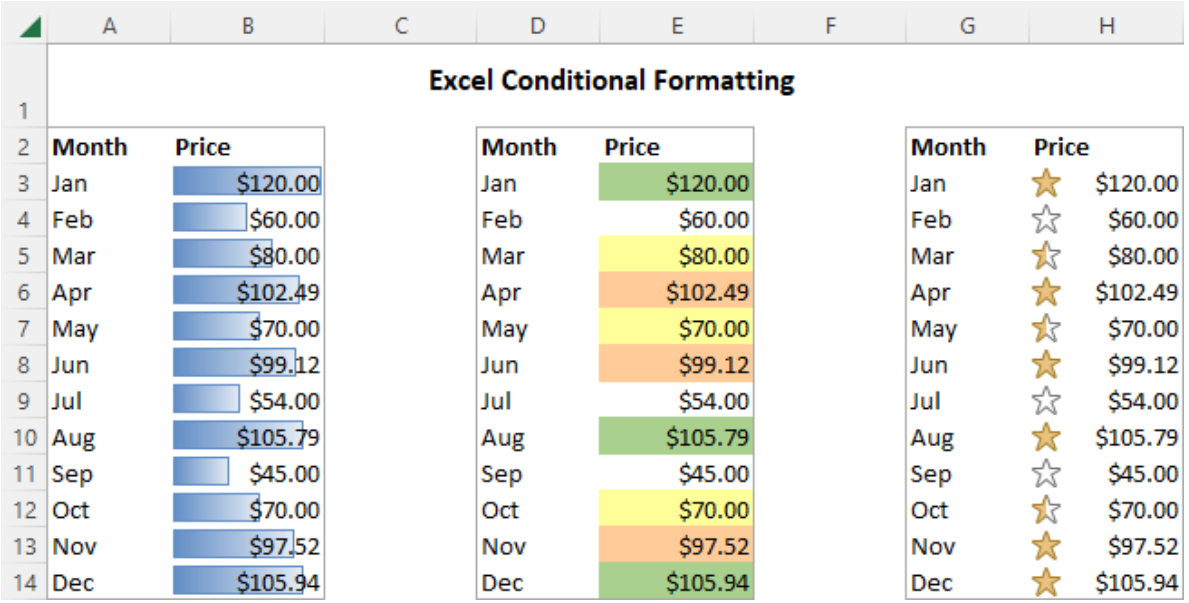
Using Referencing in Formulas

- *Example: Add a “Bonus” cell to each student’s marks using absolute referencing.*
- *Formula: =B2 + D1 (where D1 is bonus points).*
- *Now you can drag it down*



Conditional Formatting

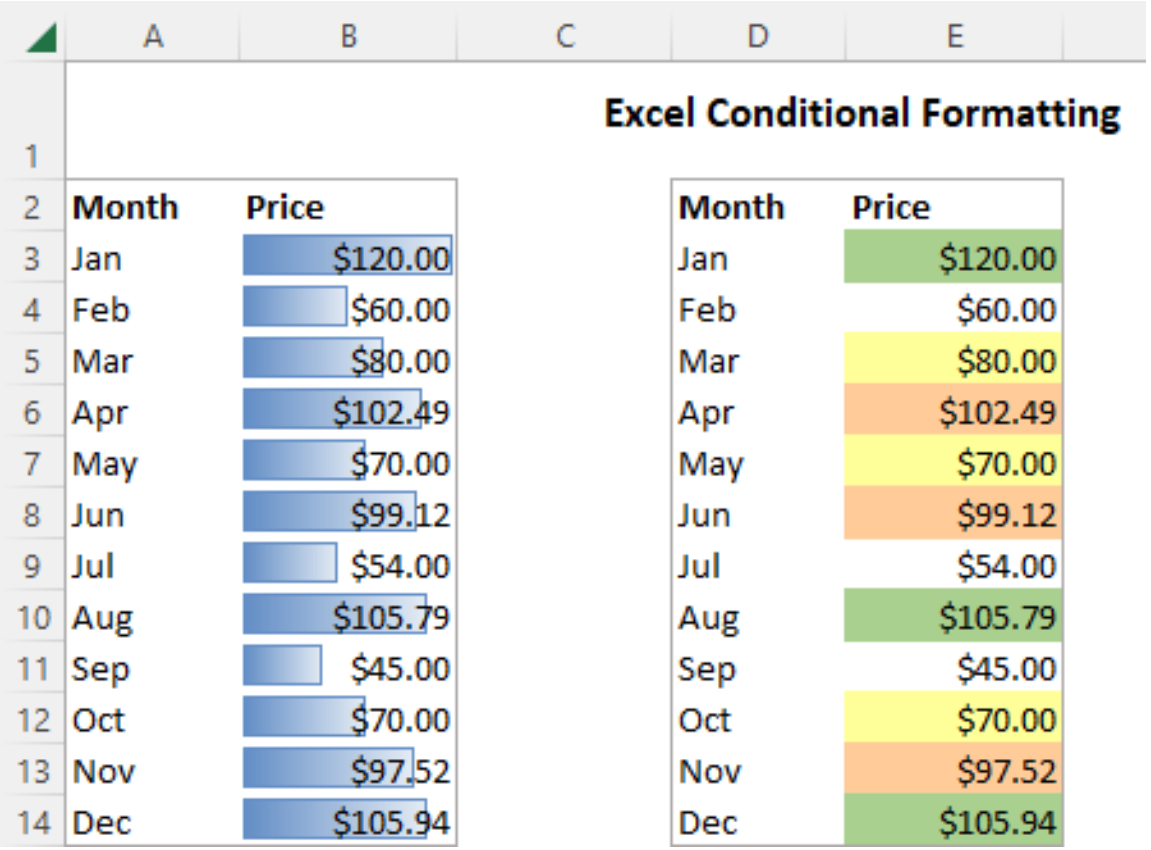
- Changes cell color based on its value.
- Highlights important numbers instantly.
- No need to draw separate charts.
- How:



Select Data → Home Tab → Conditional Formatting.

Heat Maps & Data Bars

- Data Bars: Mini bar charts inside the cell.
- The higher the number, the longer the bar.
- Color Scales: Creates a "Heat Map.
- "E.g., Green = High profit, Red = Low profit.
- Excellent for large tables of numbers.



Excel Conditional Formatting

Month	Price
Jan	$120.00
Feb	$60.00
Mar	$80.00
Apr	$102.49
May	$70.00
Jun	$99.12
Jul	$54.00
Aug	$105.79
Sep	$45.00
Oct	$70.00
Nov	$97.52
Dec	$105.94

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Using Icon Sets

- Adds small icons next to your numbers.
- Examples: Traffic lights, arrows, or stars.
- Green Arrow: Sales went up.
- Red Arrow: Sales went down.
- Makes reports look highly professional

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Creating Drop-Down Lists

- Stops users from making spelling mistakes.
- Forces them to pick from a menu (e.g., "Pass" or "Fail", "Paid" or "Unpaid").
- How to make one:

Select the empty cells.

↓

Go to the Data tab

↓

Data Validation.

↓

Change "Allow" to List.

↓

Type your options separated by commas (e.g., Pass, Fail, Merit).

Practice Task

1. *Enter the dataset in Excel.*
2. *Sort the table by any subject*

(e.g., sort by Maths High → Low).

3. *Convert the range into a table using Insert → Table.*
4. *Add Formulas:*

- *Create a column:*

  Total Marks = SUM(English, Maths, Science, History)

- *Create another column:*

  Average = AVERAGE(English, Maths, Science, History)

- *Use referencing to add a bonus mark*

  (e.g., =Total + F1 where F1 contains bonus value)

5. *Format table with header color, bold fonts.*

Save as

Student Name	English	Maths	Science	History
Nimal	78	82	85	70
Amal	65	88	74	72
Sunil	90	82	89	86
Kamal	70	75	91	75
Dilani	87	93	88	92

Recap & Next Session

You learned:

- *Sorting data*
- *Using tables*
- *Referencing in formulas*

Next: Charts in Excel (Bar, Line, Pie).

Send Me Your Thoughts (Anonymously!)

Scan the QR code and share your questions or feedback

Thank You!